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A Wavelet-Driven State-Space Architecture for Hyperspectral Unmixing via Physically-Grounded Endmember Optimization

Qiwen Jin, Ziheng Wei, Quan Liu, Jiwei Hu, Chengli Peng* and Hao Zhang*

***Abstract*—Hyperspectral unmixing (HU) is essential for sub-pixel level material characterization in remote sensing. Although deep learning-based unsupervised unmixing has shown considerable potential, most existing frameworks operate in the raw spatial-spectral domain, and are thus highly susceptible to sensor noise, spectral variability, and feature redundancy. To address these issues, this article proposes a wavelet-driven state-space architecture for robust HU. Central to the framework is the Enhanced Wavelet Transform (EWT), which integrates the Stationary Wavelet Transform (SWT) with Fourier-domain rectification to decompose the input into distinct spatial-frequency sub-bands, suppressing noise while preserving fine-grained spatial details and the per-pixel spectral profile. Based on this representation, a multi-scale spectral-spatial Mamba module captures long-range cross-band dependencies, and an abundance aggregation module fuses the complementary sub-band information for accurate abundance estimation. Equally central is a physically grounded endmember module that explicitly addresses endmember variability by learning spectral signatures through the weighted averaging of trainable prototypes and a Global Correlation Selection (GCS) mechanism. By anchoring the estimated signatures to the global spectral manifold of the scene, the mechanism keeps the learned endmembers physically interpretable and consistent with the overall data distribution. Experiments on both synthetic and real-world hyperspectral datasets show that the proposed method achieves competitive accuracy compared with existing unmixing methods in terms of both reconstruction fidelity and endmember spectral accuracy, offering a robust solution for complex remote sensing scenes. Code will be available at https://github.com/qiwenjjin/WMSM-Net.**

***Index Terms*—Hyperspectral unmixing (HU), remote sensing, deep learning, wavelet transform, state-space model (SSM).**

I. Introduction

HYPERSPECTRAL images (HSIs) acquire tens to hundreds of continuous spectral bands from visible to shortwave infrared, providing an important data source for characterizing the composition of Earth's surface materials in remote sensing. Consequently, they have been widely applied in agricultural monitoring, mineral exploration, environmental surveillance, and military reconnaissance. Their rich spectral–spatial information benefits applications such as land cover classification, change detection, target recognition, and environmental monitoring.

However, due to the trade-off between spatial and spectral resolutions and the complex distribution of ground objects, HSIs inevitably contain many mixed pixels, where a single pixel comprises multiple material components. This phenomenon restricts fine-grained analysis. To address this issue, hyperspectral unmixing (HU) [1] aims to decompose mixed pixels into spectrally pure endmembers and their abundance distributions, enabling sub-pixel characterization of material composition and distribution.

Despite substantial progress, HU remains challenging in practice. The observed HSI is often corrupted by mixed sensor noise entangled with task-relevant spatial-spectral details [2], [3], the same material can exhibit spectrally variable signatures across pixels due to illumination and atmospheric conditions [4], and nonlinear mixing effects further violate the linear assumption in complex scenes [5], [6]. Moreover, recovering physically interpretable endmembers and escaping local minima in non-convex optimization remain long-standing difficulties [7], [8]. These open problems continue to drive more accurate and physically-grounded unmixing frameworks.

To mathematically describe the mixing process, various spectral mixing models have been proposed. Among them, the linear mixing model (LMM) is widely used due to its physical interpretability and computational tractability [9]. The LMM assumes that the observed pixel spectrum is a linear combination of endmember signatures weighted by their abundances, subject to the abundance non-negativity constraint (ANC) and the abundance sum-to-one constraint (ASC). Although nonlinear mixing models (NLMM) have been proposed to model complex interactions [5], [6], the LMM remains the dominant framework for most unmixing algorithms.

A. Traditional Unmixing Methods

Traditional unmixing algorithms are generally categorized into geometrical and statistical approaches [1]. Geometrical methods exploit the simplex structure of hyperspectral data, assuming that simplex vertices correspond to pure materials (endmembers). Representative algorithms include the Pixel Purity Index (PPI) [10], N-FINDR [11], and Vertex Component Analysis (VCA) [12]. While computationally efficient, these methods rely heavily on the pure pixel assumption,

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which is often violated in real scenarios due to limited spatial resolution, leading to degraded performance in highly mixed environments. To relax this assumption, nonpure-pixel methods estimate virtual endmembers via minimum-volume simplex fitting, such as Minimum-Volume Enclosing Simplex (MVES) [13], Simplex Identification via Split Augmented Lagrangian (SISAL) [14], and Minimum Volume Simplex Analysis (MVSA) [15].

Statistical methods have also attracted considerable attention. Among them, Non-negative Matrix Factorization (NMF) [16] casts unmixing as blind source separation, jointly estimating endmembers and abundances. Because of nonconvexity and non-uniqueness, various constraints are introduced to regularize the solution, such as simplex volume minimization in MVC-NMF [7] and abundance sparsity in $L_{1/2}$-NMF [17]. Although the minimum-volume criterion is theoretically identifiable when binary mixtures lie on each simplex facet [8], in more general settings these methods, despite reducing reliance on pure pixels, remain sensitive to initialization, prone to local minima, and computationally costly.

In addition, semi-supervised approaches based on sparse regression, such as SUnSAL [3], leverage prior knowledge provided by an external spectral library (e.g., the USGS library [18]) to represent mixed pixels as sparse combinations of spectral signatures, thereby avoiding direct endmember extraction. However, spectral variability and mismatch [19], caused by factors such as illumination and atmospheric conditions, introduce discrepancies between the observed and library spectra, thereby degrading abundance estimation accuracy.

Despite significant progress, traditional methods face common limitations. Most early approaches operate in a pixel-wise manner based on spectral signatures and largely ignore spatial contextual information. Although later studies incorporated spatial context through various spatial constraint methods [20], capturing long-range spectral-spatial dependencies remains challenging. Moreover, shallow models and fixed priors limit robustness to noise and spectral variability, making it difficult to model complex spectral-spatial relationships adaptively. These limitations highlight the need for more advanced models with stronger feature extraction and denoising capabilities.

B. Deep Learning-Based Unmixing Methods

Deep learning has reshaped hyperspectral unmixing with modeling beyond traditional methods. The autoencoder (AE) has become a dominant paradigm for blind unmixing, where the encoder estimates abundances and the decoder reconstructs spectral signatures [21], [22]. However, basic AE frameworks mainly optimize reconstruction fidelity and often lack explicit physical and spatial constraints, limiting the robustness and interpretability of the results. To address this, more structured architectures have been proposed. For example, CyCU-Net enforces cycle consistency through cascaded autoencoders [23] , and two-stream autoencoders decouple spectral and spatial features via parallel branches before fusion [24], improving robustness under complex mixing conditions.

Besides AE frameworks, convolutional neural networks (CNNs) provide effective tools for modeling hyperspectral data. Among them, 3D-CNNs outperform 1D and 2D variants by jointly capturing spectral and spatial features [25]. However, CNNs are limited by local receptive fields. Although deeper architectures can enlarge the receptive field, they increase computational cost and may degrade feature resolution, making it difficult to capture long-range dependencies.

Transformers address the limitation of CNNs by employing self-attention to capture long-range spectral-spatial dependencies. Representative methods include UnDAT [26], UST-Net [27], and DeepTrans [28], which demonstrate the effectiveness of self-attention in modeling global spectral-spatial interactions. Nevertheless, self-attention incurs quadratic computational complexity with sequence length, leading to substantial memory and computational costs for hyperspectral data [29]. To alleviate this issue, hybrid CNN-Transformer architectures, such as Conformer [30], combine local convolution with global dependency modeling.

Recently, foundation models and state-space models have demonstrated remarkable potential in hyperspectral image analysis beyond hyperspectral unmixing [31]. SpectralGPT [32] leverages large-scale pretraining to learn transferable spectral-spatial representations, while MambaLG [33] employs efficient state-space modeling to capture long-range dependencies. Nevertheless, most existing methods still process hyperspectral images directly in the spatial-spectral domain, where high-frequency details and noise are entangled [34]. Consequently, effectively separating signal and noise while preserving fine spectral-spatial details and maintaining efficient global dependency modeling remains an open challenge for hyperspectral unmixing.

C. Motivation and Contributions

Existing deep learning-based unmixing frameworks primarily operate in the raw spatial–spectral domain, where effective features and noise are entangled, and conventional feature extractors lack explicit mechanisms to decouple these components, typically resulting in a trade-off between oversmoothing and noise retention. A frequency-disentangled paradigm addresses this issue by decomposing the signal into different frequency components before feature extraction. The wavelet transform is well suited for this purpose. Its multi-resolution analysis separates spatially smooth material structures in the low-frequency sub-band from fine textures and noise in the high-frequency sub-bands. The standard discrete wavelet transform (DWT) downsamples the signal at each level, destroying shift invariance and reducing the spatial resolution needed for pixel-wise abundance estimation. To preserve this resolution, we adopt the Stationary Wavelet Transform (SWT) [35], which removes the downsampling operation and preserves the original spatial resolution across all sub-bands. By further integrating a Fourier Feature Rectifier (FFR), we develop the Enhanced Wavelet Transform (EWT), enabling effective noise suppression in high-frequency sub-bands while preserving the fine textures required for accurate abundance estimation. Meanwhile, the low-frequency sub-band retains spatially coherent material structures and physically meaningful spectral signatures, providing reliable guidance for endmember generation in the dedicated endmember module.

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After feature decomposition, the resulting multi-frequency sub-bands still require a model that can capture the long-range dependencies across sub-bands. To support the EWT representation, we employ a Cascaded Spectral-Spatial Mamba (CSSMamba) module built upon the State Space Model (SSM) [36], which cascades spectral and spatial state-space modeling to progressively capture spectral correlations and spatial contextual interactions. Unlike existing Mamba-based approaches that operate on the original feature domain, CSSMamba is tailored to the frequency-disentangled sub-bands produced by EWT, so that the complementary information across frequency components is more fully exploited. Building on these sub-band features, an Abundance Aggregation Module (AAM) then fuses the complementary high- and low-frequency information in a hierarchical manner to produce accurate abundance estimates.

However, superior feature extraction does not inherently guarantee physical fidelity. Purely data-driven reconstruction often prioritizes minimizing the objective loss at the expense of spectral interpretability, leading to signatures that violate physical mixing principles [19]. To address this issue, we design an Endmember Module that learns endmember signatures through the weighted aggregation of trainable prototypical sequences. Furthermore, a Global Correlation Selection (GCS) mechanism, anchored by global spectral descriptors, is introduced to refine the estimated signatures. Through this design, the estimated endmembers are not merely numerical optima but are physically interpretable and consistent with the global spectral distribution of the scene, reducing the impact of endmember variability.

Building on these insights, we introduce a wavelet-driven state-space paradigm that rethinks feature representation and endmember optimization in HU. The main contributions of this work are summarized as follows:

1) We propose the Enhanced Wavelet Transform (EWT), which integrates the Stationary Wavelet Transform with a Fourier Feature Rectifier (FFR). By explicitly decomposing the spatial information of the HSI into complementary spatial-frequency sub-bands, EWT suppresses sensor noise while preserving task-relevant spatial textures for abundance estimation, without altering the spectral profiles.
2) Following the EWT decomposition, we propose the Abundance Extraction Module (AEM) and the Abundance Aggregation Module (AAM). Within the AEM, CSSMambaFormer performs cascaded spectral and spatial state-space modeling to capture long-range spectral-spatial dependencies. The AAM subsequently aggregates complementary structural and fine-grained detail features in a hierarchical manner, enabling spatially coherent abundance recovery.
3) We design an Endmember Module that combines prototypical sequence learning with a Global Correlation Selection (GCS) mechanism. Guided by global spectral descriptors, this module improves robustness to endmember variability while ensuring physically interpretable endmember estimation that remains consistent with the intrinsic spectral distribution of the hyperspectral scene.

The remainder of this paper is organized as follows. Section II reviews the related works. Section III details the architecture of the proposed WMSM-Net. Section IV validates the method through comprehensive experiments on synthetic and real-world datasets. Section V concludes this article.

II. Related Works

In this section, given that the proposed method builds upon established theoretical frameworks, we provide a concise overview of the pertinent prior work, including the linear mixing model (LMM), autoencoder-based architectures, and the wavelet transform.

A. AE-Based Method for LMM Unmixing

The LMM serves as the predominant theoretical foundation for hyperspectral unmixing, offering a tractable and physically intuitive approximation of complex macroscopic scattering. It simply models each observed spectrum as a linear mixture of endmembers (spectra of pure materials) scaled by their fractional coverage, alongside inevitable sensor noise [9]. Let $\mathbf{Y} \in \mathbb{R}^{L\times N}$ denote the observed hyperspectral image, comprising N pixels and L spectral bands. The unmixing process under the LMM framework is expressed as:

$$\mathbf{Y} = \mathbf{M}\mathbf{A} + \mathbf{N} \tag{1}$$

where $\mathbf{M} \in \mathbb{R}^{L\times P}$ represents the endmember matrix containing the spectral signatures of P distinct materials, $\mathbf{A} \in \mathbb{R}^{P\times N}$ denotes the corresponding abundance matrix, and $\mathbf{N} \in \mathbb{R}^{L\times N}$ accounts for inevitable system noise and measurement errors. The abundance coefficient is subject to physical constraints of nonnegativity constraint (ANC) and sum-to-one constraint (ASC) [37]:

$$\text{ANC:} \quad a_{i,j} \geq 0, \quad \forall i, j \tag{2}$$

$$\text{ASC:} \quad \sum_{i=1}^{P} a_{i,j} = 1, \quad \forall j \tag{3}$$

where $a_{i,j}$ denotes the abundance fraction of the i-th endmember at the j-th pixel. Autoencoders serve as a dominant paradigm for hyperspectral unmixing. The architecture explicitly mirrors the LMM by coupling a non-linear inference encoder with a strictly linear decoder. The encoder projects the observed hyperspectral image $\mathbf{Y} \in \mathbb{R}^{L\times N}$ into a compact latent space:

$$\mathbf{A} = f_e(\mathbf{Y}) = f(\mathbf{W}\mathbf{Y} + \mathbf{b}) \tag{4}$$

where $\mathbf{W}$ and $\mathbf{b}$ are the trainable parameters, and $f(\cdot)$ introduces non-linearity. This latent output $\mathbf{A} \in \mathbb{R}^{P\times N}$ captures the fractional abundances of P endmembers and inherently requires structural regularization to satisfy the physical ANC and ASC constraints. The decoder reconstructs the signal $\widehat{\mathbf{Y}}$ via a purely linear transformation devoid of biases:

$$\widehat{\mathbf{Y}} = f_d(\mathbf{A}) = \mathbf{M}\mathbf{A} \tag{5}$$

where the decoder weights $\mathbf{M}$, which can be regarded as the endmember matrix to be estimated, explicitly represent the P pure spectral signatures [21]. The model continuously refines both $\mathbf{A}$ and $\mathbf{M}$ by minimizing a global reconstruction penalty.

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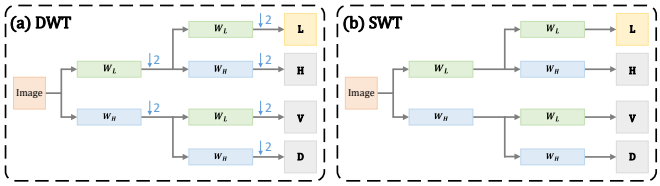


Fig. 1. (a) The process of DWT. (b) The process of SWT. $\mathbf{L}$ denotes the low-frequency signal. $\mathbf{H}$, $\mathbf{V}$, and $\mathbf{D}$ denote horizontal high-frequency signal, vertical high-frequency signal, and diagonal high-frequency signal, respectively.

B. Wavelet Transform

The Wavelet Transform (WT) provides a rigorous foundation for hyperspectral image analysis through its ability to simultaneously capture low-frequency spatial structures and high-frequency spatial details [38]. In contrast to standard convolutional filters, which often confuse random noise with fine textures, the multi-resolution analysis of WT decomposes an image into distinct frequency sub-bands, in which spatially coherent structures and fine-grained details are represented separately. The continuous wavelet transform of a signal $x(t)$ is defined as

$$W(a,b) = \frac{1}{\sqrt{|a|}} \int_{-\infty}^{+\infty} x(t)\psi^* \left(\frac{t-b}{a}\right) dt, \tag{6}$$

where $\psi(\cdot)$ denotes the mother wavelet, a represents the scale parameter controlling the frequency resolution, and b denotes the translation parameter determining the spatial location. The superscript $*$ denotes complex conjugation. As visually contrasted in Fig. 1 (a), in practical image processing, the standard DWT consists of filter banks composed of low-pass and high-pass filters. It achieves multi-scale decomposition by convolving the input image and combining it with downsampling operations. For a two-dimensional image $\mathbf{I}$ at decomposition level j, the operation produces four sub-bands:

$$\begin{aligned}
\mathbf{L}_j^{DWT} &= \left(\mathbf{L}_{j-1}^{DWT} * \mathbf{W}_L * \mathbf{W}_L\right) \downarrow 2, \\
\mathbf{H}_j^{DWT} &= \left(\mathbf{L}_{j-1}^{DWT} * \mathbf{W}_L * \mathbf{W}_H\right) \downarrow 2, \\
\mathbf{V}_j^{DWT} &= \left(\mathbf{L}_{j-1}^{DWT} * \mathbf{W}_H * \mathbf{W}_L\right) \downarrow 2, \\
\mathbf{D}_j^{DWT} &= \left(\mathbf{L}_{j-1}^{DWT} * \mathbf{W}_H * \mathbf{W}_H\right) \downarrow 2,
\end{aligned} \tag{7}$$

where $\mathbf{W}_L$ and $\mathbf{W}_H$ denote the low-pass and high-pass filters, respectively, and $\downarrow$ 2 indicates the downsampling operation. While computationally efficient, the downsampling operation destroys the shift-invariance of the signal, inevitably leading to spatial information loss, structural distortion, and jagged artifacts at object boundaries. Such degradation severely hinders the network's ability to accurately estimate pixel-wise fractional abundances. As depicted in Fig. 1 (b), the SWT achieves shift-invariance by replacing the downsampling operation with upsampled filter coefficients. By doing so, it ensures that every decomposed frequency sub-band strictly maintains the identical spatial dimensions and resolution as the original input image. Let $\mathbf{W}_L^{(j)}$ and $\mathbf{W}_H^{(j)}$ be the filters at level j, derived by inserting $2^{j-1}-1$ zeros between consecutive coefficients of the original filters. The SWT decomposition is expressed as:

$$\begin{aligned}
\mathbf{L}_j^{SWT} &= \mathbf{L}_{j-1}^{SWT} * \mathbf{W}_L^{(j)} * \mathbf{W}_L^{(j)} \\
\mathbf{H}_j^{SWT} &= \mathbf{L}_{j-1}^{SWT} * \mathbf{W}_L^{(j)} * \mathbf{W}_H^{(j)} \\
\mathbf{V}_j^{SWT} &= \mathbf{L}_{j-1}^{SWT} * \mathbf{W}_H^{(j)} * \mathbf{W}_L^{(j)} \\
\mathbf{D}_j^{SWT} &= \mathbf{L}_{j-1}^{SWT} * \mathbf{W}_H^{(j)} * \mathbf{W}_H^{(j)}
\end{aligned} \tag{8}$$

By maintaining the original spatial resolution across all frequency sub-bands, SWT ensures structural consistency and provides a more reliable feature set for hyperspectral unmixing [39].

III. Proposed Method

In this section, we introduce the architecture of the proposed WMSM-Net, as illustrated in Fig. 2. Formulated within a deep autoencoder framework adhering to the LMM, the architecture comprises two primary components. The abundance module consists of three modules, which are the enhanced wavelet transform (EWT), the abundance extraction module (AEM), and the abundance aggregation module (AAM). The endmember module utilizes the global spectral feature extraction module (GSFEM) to guide the weighted average endmember extraction module (WAEEM) for extracting endmembers. Finally, we discuss the loss function.

A. Enhanced Wavelet Transform (EWT)

Hyperspectral images provide complementary spectral information and spatial structures. The spatial structures exhibit distinct spatial frequency characteristics, making effective extraction of informative spatial features while preserving spatial resolution a key challenge in hyperspectral unmixing. We address this via the EWT, a two-stage pipeline combining the SWT and FFR. As illustrated in Fig. 3, the SWT removes the aliasing introduced by the downsampling of the standard DWT, and the subsequent FFR further suppresses the residual high-frequency noise in homogeneous regions while preserving the structural edges, yielding a cleaner frequency-disentangled representation.

1) Stationary Wavelet Transform (SWT): The standard DWT downsamples the input signal, which destroys the fine spatial details essential for accurate pixel-wise abundance estimation. We bypass this limitation by employing the SWT. Let $\mathbf{X}$ denote the input HSI. The SWT decomposes $\mathbf{X}$ into four distinct frequency sub-bands:

$$\mathrm{SWT}(\mathbf{X}) = [\mathbf{X}_{LF}, \mathbf{X}_{HF}, \mathbf{X}_{VF}, \mathbf{X}_{DF}] \tag{9}$$

where $\mathbf{X}_{LF}$ represents the low-frequency approximation containing coarse-grained spectral consistency. $\mathbf{X}_{HF}$, $\mathbf{X}_{VF}$, and $\mathbf{X}_{DF}$ denote the high-frequency details corresponding to horizontal, vertical, and diagonal spatial textures, respectively.

2) Fourier Feature Rectifier (FFR): The decomposed coefficients from the SWT inherently retain task-irrelevant noise. We introduce the FFR (Fig. 4) to clean these representations by leveraging global dependencies. Given a wavelet feature map $\mathbf{F}_{in}$, we employ the Fast Fourier Transform (FFT) to convert

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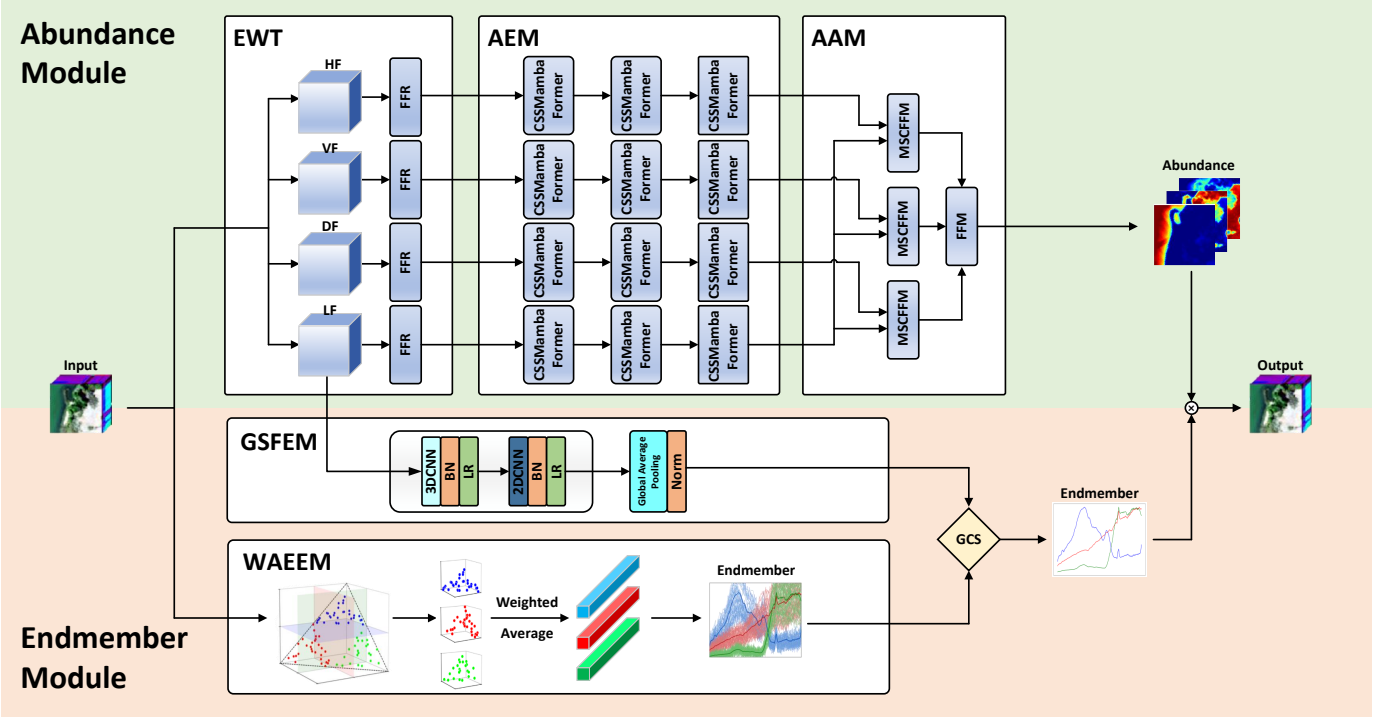


Fig. 2. The overall architecture of the proposed wavelet-enhanced multi-scale spectral-spatial mamba network (WMSM-Net).

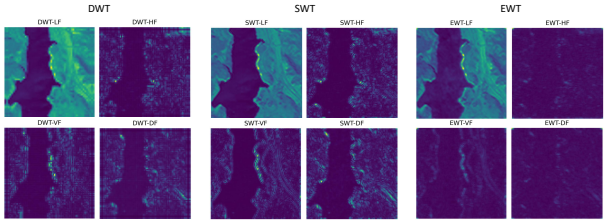


Fig. 3. Visualization of the DWT, SWT, and EWT wavelet sub-bands on the Jasper Ridge dataset.

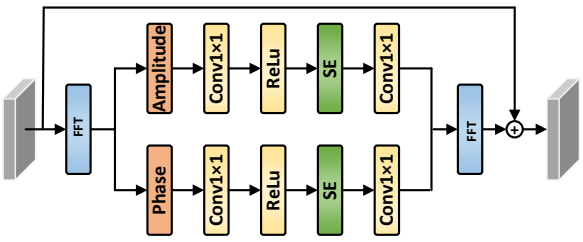


Fig. 4. Schematic of the fourier feature rectifier (FFR).

spatial features into the frequency domain. The transform function $\mathcal{F}$ which converts $\mathbf{F}_{in}$ to the Fourier space $\mathbf{Z}$ can be expressed as:

$$\begin{aligned}\mathcal{F}(\mathbf{F}_{in})(u,v) &= \frac{1}{\sqrt{HW}}\sum_{h=0}^{H-1}\sum_{w=0}^{W-1}\mathbf{F}_{in}(h,w)e^{-j2\pi(\frac{hu}{H}+\frac{wv}{W})} \\ &= \mathbf{Z}(u,v)\end{aligned} \tag{10}$$

where h, w are the coordinates in the spatial space and u, v are the coordinates in the Fourier space. $\mathbf{Z}(u,v)$ can be

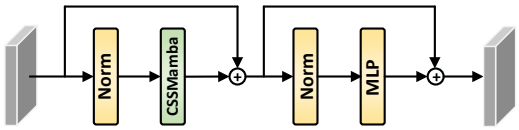


Fig. 5. Schematic of the CSSMambaFormer.

decomposed as:

$$\mathbf{Z}(u,v) = \mathcal{R}(\mathbf{Z}(u,v)) + j\mathcal{I}(\mathbf{Z}(u,v)) \tag{11}$$

where $\mathcal{R}(\cdot)$ and $\mathcal{I}(\cdot)$ represent the real and imaginary units, respectively. This enables the network to handle $\mathcal{R}(\cdot)$ and $\mathcal{I}(\cdot)$ units independently. To dynamically recalibrate these frequency responses, we implement the dual-branch structure depicted in Fig. 4. Each branch feeds the respective component through a 1×1 convolution, a ReLU activation, and an SE block. This rectification process is formulated as:

$$\begin{aligned}\mathbf{Z}'_{\mathcal{P}} &= \mathrm{Conv}_{1\times1}(\mathrm{SE}(\mathrm{ReLU}(\mathrm{Conv}_{1\times1}(\mathcal{P}(\mathbf{Z}))))) \\ &\mathcal{P} \in \{\mathcal{R}, \mathcal{I}\}\end{aligned} \tag{12}$$

where the rectified spectrum is reconstructed by $\mathbf{Z}' = \mathbf{Z}'_{\mathcal{R}} + j\mathbf{Z}'_{\mathcal{I}}$. Finally, the Inverse FFT (IFFT) is applied to map the features back to the spatial domain, followed by a residual connection to yield the final output $\mathbf{F}_{out}$:

$$\mathbf{F}_{out} = \mathbf{F}_{in} + \mathcal{F}^{-1}(\mathbf{Z}') \tag{13}$$

B. Abundance Extraction Module (AEM)

The AEM serves as the backbone of the unmixing network, processing the HF, VF, DF, and LF components through four structurally identical parallel branches. Each branch

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achieves deep feature extraction by stacking multiple CSS-MambaFormer blocks. As illustrated in Fig. 5, the forward propagation of the i-th CSSMambaFormer block is formulated as:

$$\begin{aligned} \mathbf{F}'_i &= \text{CSSMamba}(\text{LN}(\mathbf{F}_{i-1})) + \mathbf{F}_{i-1} \\ \mathbf{F}_i &= \text{MLP}(\text{LN}(\mathbf{F}'_i)) + \mathbf{F}'_i \end{aligned} \tag{14}$$

where $\mathbf{F}_{i-1}$, $\mathbf{F}_i$, and $\mathbf{F}'_i$ represent the input, output, and intermediate features, respectively. LN, CSSMamba, and MLP represent layer normalization, the cascaded spectral-spatial mamba, and multi-layer perceptron, respectively. The core component of the SSEM, the CSSMamba, disentangles spatial-spectral correlations through a cascaded dual-branch architecture (Fig. 6). The spectral branch first extracts global spectral consistency by processing downsampled features through a Mamba block and a Dynamic Transformation (DyT) layer, restoring resolution via upsampling and a residual connection. The spectral branch can be represented as:

$$\begin{aligned} \mathbf{T}_{spe1} &= \text{Mamba}\left(\text{Down}(\mathbf{F}_{in})\right) \\ \mathbf{T}_{spe2} &= \text{Up}\left(\sigma\left(\text{DyT}\left(\mathbf{T}_{spe1}\right)\right)\right) \\ \mathbf{F}_{spe} &= \mathbf{F}_{in} + \mathbf{T}_{spe2} \end{aligned} \tag{15}$$

where $\mathbf{F}_{in}$, $\mathbf{T}_{spe1}$, $\mathbf{T}_{spe2}$, and $\mathbf{F}_{spe}$ represent the input, the intermediate spectral sequence features captured by the Mamba block, the refined spectral features restored to the original spatial resolution, and the output of the spectral branch, respectively. $\text{Down}(\cdot)$, $\text{Up}(\cdot)$, $\text{DyT}(\cdot)$, and $\sigma(\cdot)$ denote downsampling, upsampling, dynamic transformation layer, and SiLU activation. Subsequently, the spatial branch utilizes the spectrally enhanced features $\mathbf{F}_{spe}$ to model long-range dependencies. The spatial branch can be represented as:

$$\begin{aligned} \mathbf{T}_{spa1} &= \text{Mamba}\left(\text{MSPAM}(\mathbf{F}_{spe})\right) \\ \mathbf{T}_{spa2} &= \text{Reshape}\left(\sigma\left(\text{DyT}\left(\mathbf{T}_{spa1}\right)\right)\right) \\ \mathbf{F}_{spa} &= \mathbf{F}_{in} + \mathbf{T}_{spa2} \end{aligned} \tag{16}$$

where $\mathbf{T}_{spa1}$, $\mathbf{T}_{spa2}$, and $\mathbf{F}_{spa}$ denote the intermediate spatial dependency features modeled by the Mamba block, the recalibrated spatial features after dynamic transformation and reshaping, and the output of the spatial branch, respectively. $\text{MSPAM}(\cdot)$ represents the multi-scale pyramid aggregation module for multi-scale spatial context aggregation. The decoupled features from the spatial and spectral branches are integrated via element-wise summation, yielding the initial fused feature map $\mathbf{F}_{fuse}$:

$$\mathbf{F}_{fuse} = \mathbf{F}_{spa} + \mathbf{F}_{spe} \tag{17}$$

The fused features are further refined by the global-local collaborative attention (GLCA) module, which captures both global context and local details through two parallel pathways. The global pathway models channel dependencies via spatial average pooling, while the local pathway focuses on high-frequency responses using channel-wise maximum pooling. Both pathways employ the dual-weight expansion convolution (DWEConv) followed by a Sigmoid activation to generate attention maps:

$$\begin{aligned} \mathbf{A}_{global} &= \sigma\left(\text{DWEConv}\left(\text{AvgPool}(\mathbf{F}_{fuse})\right)\right) \\ \mathbf{A}_{local} &= \sigma\left(\text{DWEConv}\left(\text{MaxPool}_c(\mathbf{F}_{fuse} * \mathbf{A}_{global})\right)\right) \end{aligned} \tag{18}$$

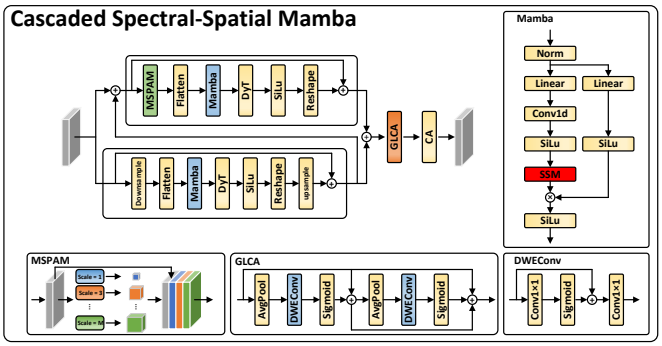


Fig. 6. Schematic of the cascaded spectral-spatial mamba (CSSMamba).

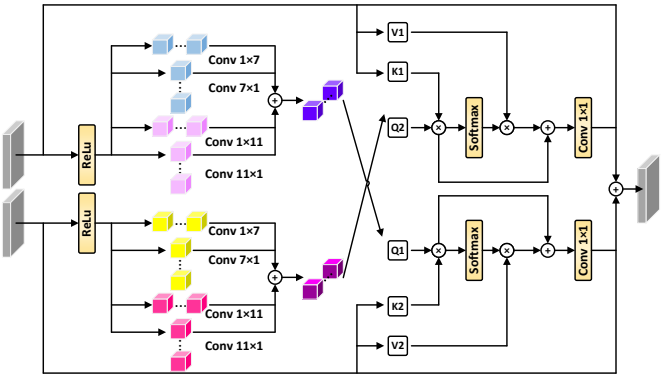


Fig. 7. Schematic of multi-scale cross-frequency fusion module (MSCFFM).

where $\mathbf{A}_{global}$ and $\mathbf{A}_{local}$ denote the attention weights from the global and local pathways, respectively. $\text{AvgPool}(\cdot)$ and $\text{MaxPool}_c(\cdot)$ represent spatial average pooling and channel-wise maximum pooling [40]. The final calibrated feature map is obtained by dynamically aggregating weights:

$$\mathbf{F}_{GLCA} = \mathbf{F}_{fuse} * \mathbf{A}_{global} * \mathbf{A}_{local} \tag{19}$$

Finally, a channel attention (CA) is applied to $\mathbf{F}_{GLCA}$ to further refine channel dependencies, producing the final output.

C. Abundance Aggregation Module (AAM)

The four parallel encoder branches yield distinct frequency-domain features: the low-frequency (LF) component capturing spectral consistency, and high-frequency (HF, VF, DF) components extracting fine-grained textures. To integrate these without ignoring their intrinsic semantic relationships, the abundance module utilizes the AAM. The AAM hierarchically fuses features using the multi-scale cross-frequency fusion module (MSCFFM), as illustrated in Fig. 7.

Given two distinct frequency feature maps $\mathbf{F}_1$ and $\mathbf{F}_2$, the MSCFFH first employs parallel asymmetric convolutions (kernel sizes of 1×7, 7×1, 1×11, and 11×1) to extract contextual information while maintaining computational efficiency. The multi-scale context feature $\mathbf{F}_{ms}^{(i)}$ for the i-th branch is computed as:

$$\mathbf{F}_{ms}^{(i)} = \sum_{k \in \{7,11\}} \left(\text{Conv}_{1\times k}(\mathbf{F}_i) + \text{Conv}_{k\times 1}(\mathbf{F}_i)\right), \; i \in \{1,2\} \tag{20}$$

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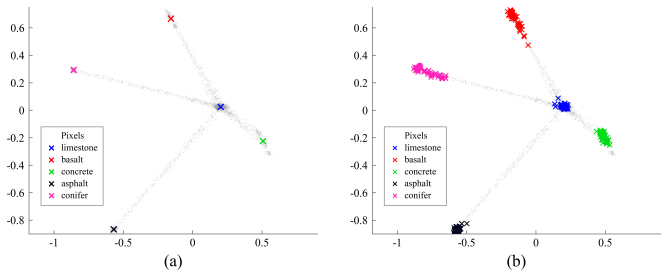


Fig. 8. Scatter plot of the synthetic dataset, where the gray dots denote the projected pixels by PCA. (a) Endmember scatter plot of the ground truth. (b) The K learnable prototypes per endmember (colored crosses), their weighted average (solid dots), and the weighted endmembers in Eq. (22).

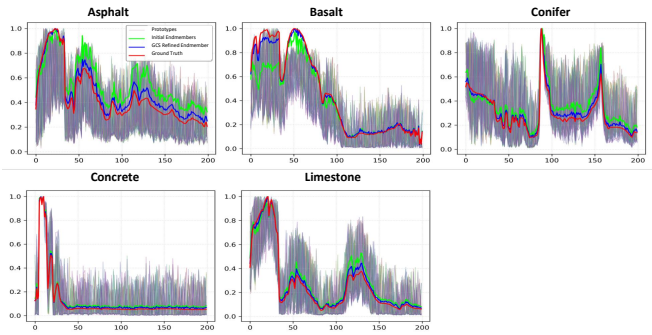


Fig. 9. Visual illustration of the endmember generation process on the simulated dataset with 10 dB SNR. Colorful shaded areas represent the K prototypes; green, blue, and red lines denote the weighted initial endmembers, the GCS refined endmember, and the ground truth, respectively.

Subsequently, collaborative feature selection is performed via a dual-path cross-attention mechanism. $\mathbf{F}_{ms}^{(1)}$ and $\mathbf{F}_{ms}^{(2)}$ are projected into Query ($\mathbf{Q}$), Key ($\mathbf{K}$), and Value ($\mathbf{V}$) tensors. The model captures reciprocal dependencies by allowing the Query of one branch to interact with the Key and Value of the opposite branch. The cross-attended representations $\mathbf{Y}_1$ and $\mathbf{Y}_2$ are formulated as:

$$\begin{aligned}\mathbf{Y}_1 &= \text{Softmax}\left(\frac{\mathbf{Q}_2\mathbf{K}_1^\top}{\sqrt{d}}\right)\mathbf{V}_1 \\ \mathbf{Y}_2 &= \text{Softmax}\left(\frac{\mathbf{Q}_1\mathbf{K}_2^\top}{\sqrt{d}}\right)\mathbf{V}_2\end{aligned} \tag{21}$$

where d serves as the scaling factor, $\mathbf{Q}_1, \mathbf{K}_1, \mathbf{V}_1$ are derived from $\mathbf{F}_{ms}^{(1)}$, and $\mathbf{Q}_2, \mathbf{K}_2, \mathbf{V}_2$ from $\mathbf{F}_{ms}^{(2)}$. The attended features $\mathbf{Y}_1$ and $\mathbf{Y}_2$ are fused via element-wise summation and refined by a 1×1 convolution to produce the MSCFFM output $\mathbf{F}_{out}$. The AAM deploys three parallel MSCFFM blocks. Using the spectrally rich LF component as the primary anchor, learnable scalar weights are applied to adaptively balance branch contributions. The weighted LF feature then independently interacts with the HF, VF, and DF components. Finally, the outputs of these three fusion heads are concatenated along the channel dimension and refined by a final fusion module (FFM) through a non-linear convolutional sequence, yielding the comprehensive feature map for abundance estimation.

D. Endmember Module

We propose an endmember module that employs an ensemble-based generation strategy. It coordinates the Weighted Average endmember extraction module (WAEEM) with the global spectral feature extraction Module (GSFEM), dynamically connecting them through a global correlation selection (GCS) mechanism to ensure that the estimated endmembers are consistent with the inherent global spectral properties. The evolution of this process is visualized in Fig. 9. As implemented in our model, we initialize a learnable query dictionary $\mathbf{Q} \in \mathbb{R}^{P\times K\times L}$, where P is the number of endmembers, K is the number of learnable prototypes per endmember, and L is the number of bands. The query vector is split according to the number of endmembers, the weights are expanded and multiplied by the query vector to obtain the weighted initial endmembers. The candidate spectral signature $\mathbf{E}_{cand}^{(p)}$ for the p-th endmember is derived by aggregating these prototypes:

$$\mathbf{E}_{cand}^{(p)} = \frac{1}{K}\sum_{k=1}^{K}\mathbf{W}_{p,k}\mathbf{Q}_{p,k} \tag{22}$$

where $\mathbf{W} \in \mathbb{R}^{P\times K}$ represents the learnable weights that dynamically adjust the contribution of each prototype. This weighted averaging mechanism stabilizes the initialization and improves the robustness against noise, as illustrated in Fig. 8. To ensure the estimated endmembers align with the intrinsic spectral properties of the scene and to mitigate spectral variability at the endmember level, we design the GSFEM to extract a global spectral feature. The Low-Frequency (LF) feature map $\mathbf{X}_{LF}$ produced by the EWT serves as the input, which retains the complete spectral profile. By capturing inter-band correlations through a cascaded 3D-2D convolutional architecture, global average pooling (GAP) and ℓ_2 normalization are applied to yield the global spectral feature $\mathbf{z}_g \in \mathbb{R}^L$:

$$\mathbf{z}_g = \text{Norm}\left(\text{GAP}\left(\text{Conv2d}\left(\mathcal{R}\left(\text{Conv3d}(\mathbf{X}_{LF}\right)\right)\right)\right)\right) \tag{23}$$

where $\mathcal{R}$ flattens the spectral-depth dimensions. The GCS mechanism acts as a spectral gate to refine the prototypes. It projects $\mathbf{E}_{cand}$ onto the global descriptor $\mathbf{z}_g$ to derive the correlation response $\mathbf{\Phi}$, which is subsequently normalized to obtain the correlation matrix $\mathbf{G}$ that measures spectral relevance. This matrix is used to modulate the candidate endmembers:

$$\begin{aligned}\mathbf{\Phi} &= \mathbf{E}_{cand} \odot \mathbf{z}_g^\top \\ \mathbf{M} &= \mathbf{E}_{cand} \odot (1 + \lambda\mathbf{G})\end{aligned} \tag{24}$$

where $\mathbf{G}$ represents the filtered correlation weights derived from $\mathbf{\Phi}$, and λ is a learnable scaling factor that controls the strength of the global correlation refinement. This interaction guides the aggregated endmembers toward the global spectral distribution of the input scene, improving their consistency with the intrinsic spectral characteristics and enhancing robustness to spectral variability.

E. Loss Function

WMSM-Net is trained using a multi-objective loss that balances reconstruction fidelity and physical priors. Reconstruction fidelity is enforced by the mean squared error (MSE)

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loss, which minimizes the pixel-wise discrepancy between the original HSI $\mathbf{Y}$ and the reconstruction $\widehat{\mathbf{Y}} = \mathbf{MA}$:

$$\mathcal{L}_{mse} = \frac{1}{N}\|\mathbf{Y} - \widehat{\mathbf{Y}}\|_F^2 \tag{25}$$

where N denotes the number of pixels. To complement the MSE, the spectral angle distance (SAD) loss is incorporated to measure the spectral shape similarity between the original and reconstructed signatures:

$$\mathcal{L}_{sad} = \frac{1}{N}\sum_{i=1}^{N} \arccos\left(\frac{\mathbf{y}_i^\top \widehat{\mathbf{y}}_i}{\|\mathbf{y}_i\|_2\|\widehat{\mathbf{y}}_i\|_2 + \epsilon}\right) \tag{26}$$

where ϵ is a small constant for numerical stability. The endmember consistency constraint aligns the mean of the estimated endmembers $\mathbf{M}$ with the global mean spectrum $\bar{\mathbf{y}}$ [24]:

$$\mathcal{L}_{em} = \|\text{Mean}(\mathbf{M}) - \bar{\mathbf{y}}\|_2^2 \tag{27}$$

Sparsity of the abundance matrix $\mathbf{A}$ is encouraged via the $\ell_{1/2}$ norm regularization [17]:

$$\mathcal{L}_{spa} = \frac{1}{N}\sum_{i=1}^{N} \|\mathbf{a}_i\|_{1/2} \tag{28}$$

The total objective function is formulated as the weighted sum of the aforementioned terms:

$$\mathcal{L}_{total} = \mathcal{L}_{mse} + \lambda_{sad}\mathcal{L}_{sad} + \lambda_{em}\mathcal{L}_{em} + \lambda_{spa}\mathcal{L}_{spa} \tag{29}$$

where λ_{sad}, λ_{em}, and λ_{spa} are hyperparameters balancing the contribution of each constraint. λ_{sad}, λ_{em}, and λ_{spa} are set to 1, 1, and 10^{-2}, respectively.

IV. Experiments

This section presents comparative experiments on synthetic datasets and three real-world hyperspectral datasets to validate the effectiveness and robustness of the proposed WMSM-Net. Eleven algorithms are used for benchmarking, including three model-based methods and eight representative deep learning models. The fully constrained least squares unmixing (FCLSU) method [37] is adopted as a representative traditional optimization-based approach, and two extended linear mixing model methods are further included, namely the extended linear mixing model (ELMM) [41] and the generalized linear mixing model (GLMM) [42]. In addition, eight recent deep learning methods are selected to cover different architectural paradigms. CNN and autoencoder-based models include CyCU [23], DFFN [43], A2SAN [44], SSAF [45], and SWC-Net [39]. Transformer-based methods are represented by DeepTrans [28] and SRViT [46]. UNMamba [47] is included to represent Mamba models.

A. Data Description

Experiments are conducted on one synthetic dataset and three real-world hyperspectral datasets. Their visualizations are shown in Fig. 10, and detailed descriptions are provided below.

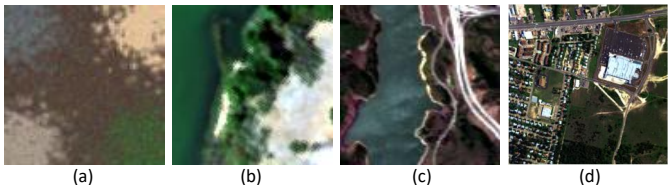


Fig. 10. The hyperspectral datasets used in the experiments. (a) Synthetic. (b) Samson. (c) Jasper Ridge. (d) Urban.

TABLE I
Network Parameter Value Setting

Dataset	Epochs	LR	Dropout	λ_{sad}	λ_{em}	λ_{spa}
Synthetic	800	0.05	0.03	1	1	10^{-2}
Samson	400	0.003	0.06	1	1	10^{-2}
Jasper Ridge	300	0.003	0.03	1	1	10^{-2}
Urban	200	0.003	0.03	1	1	10^{-2}

1) Synthetic Dataset: Generated following the protocol in TANet [48], this dataset simulates spectral variability with a spatial size of 60×60 pixels. Five endmembers (limestone, basalt, concrete, conifer, and asphalt) were selected from the ASTER library [49]. Abundances were generated using a Dirichlet distribution under the ASC and ANC constraints, with Gaussian noise added to simulate sensor degradation.

2) Samson Dataset: Acquired by the SAMSON sensor, the original image contains 156 spectral bands, with the spectral wavelengths ranging from approximately 400.94 to 899.07 nm. A sub-image of size 95×95 pixels is used for experiments, which contains three primary endmembers: soil, tree, and water.

3) Jasper Ridge Dataset: Captured by the AVIRIS sensor, the original dataset covers the visible-to-shortwave-infrared spectral region. After removing bands affected by water vapor and atmospheric effects, 198 bands remain, with the spectral wavelengths ranging from approximately 399.37 to 2457.24 nm. A region of interest sized 100×100 pixels is selected, comprising four endmembers: road, soil, water, and tree.

4) Urban Dataset: Collected by the HYDICE sensor, the Urban image has a spatial dimension of 307×307 pixels. After discarding noisy and water vapor absorption bands, 162 valid bands are retained, with the spectral wavelengths ranging from approximately 412 to 2390 nm. The scene contains four prominent endmembers: asphalt, grass, tree, and roof.

B. Evaluation Metrics

To quantitatively evaluate the performance of the proposed method, we utilize five quantitative metrics, covering both endmember extraction and abundance estimation. Endmember extraction is evaluated by SAD [9] and SID, and abundance estimation by RMSE, AAD, and AID. For all five metrics, a smaller value indicates better unmixing performance. The five metrics are defined as

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TABLE II
QUANTITATIVE PERFORMANCE COMPARISON (SAD AND RMSE) ON THE SYNTHETIC DATASET UNDER DIFFERENT NOISE LEVELS. THE BEST RESULTS ARE SHOWN IN **BOLD**, AND THE SECOND-BEST RESULTS ARE UNDERLINED.

Methods		FCLSU	ELMM	GLMM	CyCU	DeepTrans	DFFN	A2SAN	UNMamba	SSAF	SRViT	SWC-Net	WMSM-Net
SAD	10 dB	0.2575	0.2699	0.2264	0.0866	0.0721	0.1559	0.0402	0.0402	0.0558	0.0336	0.0346	**0.0175**
	20 dB	0.1020	0.0983	0.0780	0.0683	0.0270	0.1388	0.0486	0.0374	0.0391	0.0195	0.0206	**0.0170**
	30 dB	0.0371	0.0375	0.0363	0.0729	0.0221	0.0697	0.0384	0.0287	0.0340	0.0172	0.0183	**0.0132**
	40 dB	0.0332	0.0334	0.0251	0.0634	0.0214	0.0609	0.0207	0.0268	0.0327	0.0164	0.0103	**0.0060**
RMSE	10 dB	0.1756	0.1912	0.1806	0.2949	0.1260	0.2662	**0.0741**	0.1394	0.1330	0.2098	0.1031	0.0773
	20 dB	0.0812	0.0858	0.0744	0.3881	0.0667	0.2196	0.0657	0.1246	0.0700	0.2126	0.0862	**0.0619**
	30 dB	0.0513	0.0537	0.0510	0.3390	0.0579	0.1775	0.0585	0.0981	**0.0472**	0.2304	0.0726	0.0598
	40 dB	0.0501	0.0505	0.0502	0.4229	0.0570	0.1082	0.0528	0.0901	0.0455	0.2559	0.0617	**0.0313**

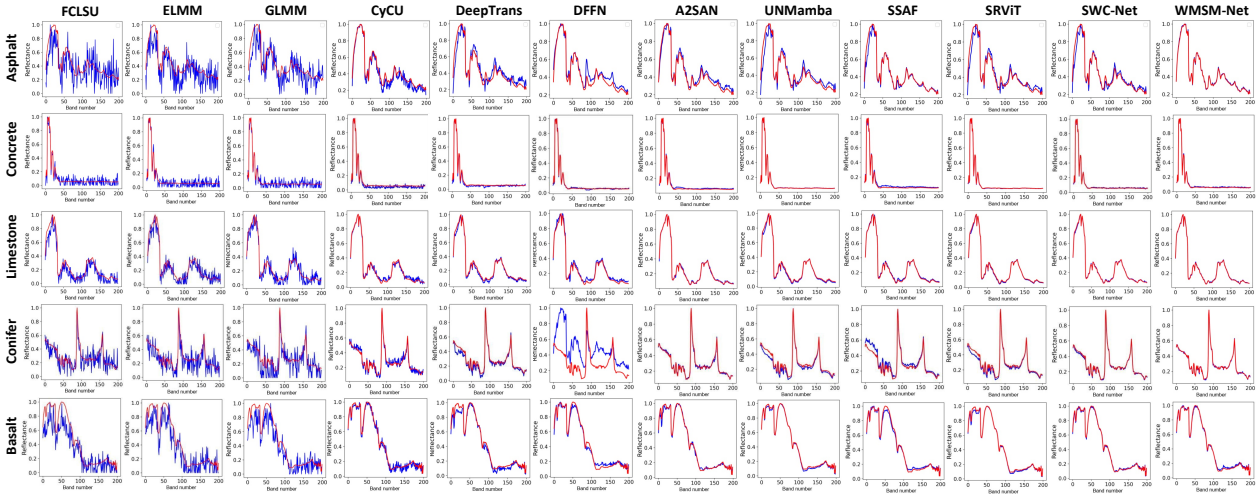


Fig. 11. Visual comparison of the endmembers extracted by different methods on the synthetic dataset under severe noise conditions (SNR = 10 dB). Red lines denote the reference spectra, and blue lines denote the estimated spectra. The x-axis represents the spectral band number, and the y-axis represents the normalized reflectance.

$$
\begin{cases}
\text{SAD} = \dfrac{1}{P}\displaystyle\sum_{k=1}^{P} \arccos\left(\dfrac{\mathbf{e}_k^\top \hat{\mathbf{e}}_k}{\|\mathbf{e}_k\|_2 \|\hat{\mathbf{e}}_k\|_2}\right) \\
\text{SID} = \dfrac{1}{P}\displaystyle\sum_{k=1}^{P} [D(\mathbf{e}_k \,\|\, \hat{\mathbf{e}}_k) + D(\hat{\mathbf{e}}_k \,\|\, \mathbf{e}_k)] \\
\text{RMSE} = \sqrt{\dfrac{1}{NP}\displaystyle\sum_{i=1}^{N}\sum_{k=1}^{P}(a_{i,k} - \hat{a}_{i,k})^2} \\
\text{AAD} = \dfrac{1}{N}\displaystyle\sum_{i=1}^{N} \arccos\left(\dfrac{\mathbf{a}_i^\top \hat{\mathbf{a}}_i}{\|\mathbf{a}_i\|_2 \|\hat{\mathbf{a}}_i\|_2}\right) \\
\text{AID} = \dfrac{1}{N}\displaystyle\sum_{i=1}^{N} [D(\mathbf{a}_i \,\|\, \hat{\mathbf{a}}_i) + D(\hat{\mathbf{a}}_i \,\|\, \mathbf{a}_i)]
\end{cases}
\tag{30}
$$

where $\mathbf{e}_k, \hat{\mathbf{e}}_k \in \mathbb{R}^L$ are the reference and estimated spectra of the k-th endmember, $\mathbf{a}_i, \hat{\mathbf{a}}_i \in \mathbb{R}^P$ are the reference and estimated abundance vectors of the i-th pixel, $a_{i,k}$ is the k-th entry of $\mathbf{a}_i$, and L, P, N are the numbers of bands, endmembers, and pixels. In our experiments, the proposed model is implemented in Python 3.9 with PyTorch 2.2.2. Detailed network parameter settings are summarized in Table I. All metrics are evaluated on a single NVIDIA GeForce RTX 4090 GPU.

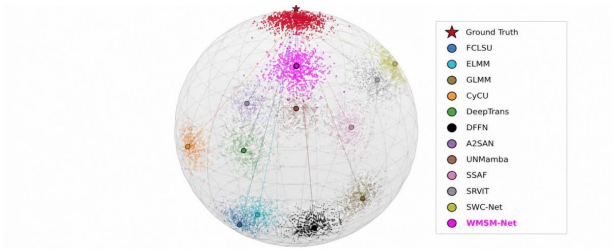


Fig. 12. Spherical visualization of spectral angle distance (SAD) distributions for different unmixing methods on the synthetic dataset under severe noise conditions (SNR = 10 dB). Each endmember lies on a unit hypersphere, and markers closer to the ground truth indicate higher accuracy.

C. Experiments on the Synthetic Dataset

To evaluate the unmixing efficiency and noise resilience of the proposed WMSM-Net, experiments were conducted on the synthetic dataset with white Gaussian noise added at signal-to-noise ratios (SNRs) of 10, 20, 30, and 40 dB. We benchmarked our method against eleven comparison algorithms, evaluating performance via SAD and RMSE, as detailed in Table II. Across all noise intensities, WMSM-Net ranked first in SAD, indicating its capability in reconstructing endmember spectra. We visualize the endmember signatures in Fig. 11 and the

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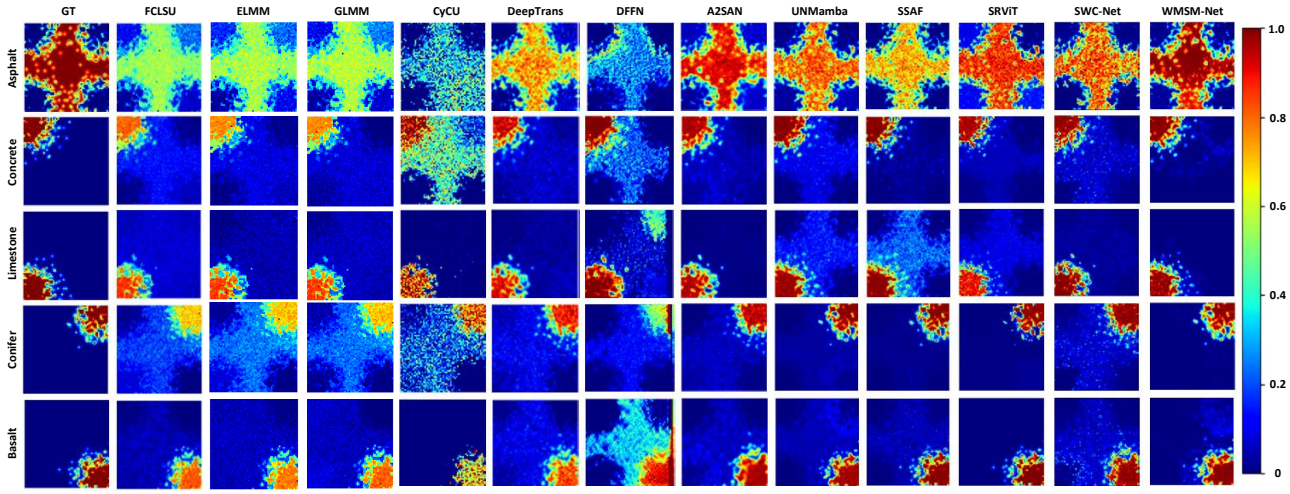


Fig. 13. Visual comparison of fractional abundance maps estimated by different unmixing methods on the synthetic dataset under severe noise conditions (SNR = 10 dB).

TABLE III
QUANTITATIVE PERFORMANCE COMPARISON (SAD, RMSE, SID, AAD, AND AID) ON THE SAMSON DATASET. THE BEST RESULTS ARE SHOWN IN **BOLD**, AND THE SECOND-BEST RESULTS ARE UNDERLINED.

Methods		FCLSU	ELMM	GLMM	CyCU	DeepTrans	DFFN	A2SAN	UNMamba	SSAF	SRViT	SWC-Net	**WMSM-Net**
SAD	Soil	0.0404	0.0404	0.0345	0.0380	**0.0100**	0.0454	0.0180	0.0354	0.0190	0.1174	0.0146	0.0108
	Tree	**0.0219**	**0.0219**	0.0262	0.0364	0.0391	0.0490	0.0388	0.0342	0.0404	0.1049	0.0310	0.0300
	Water	0.1100	0.1100	0.2049	0.1414	0.0718	0.0368	0.1210	0.0315	0.0650	0.1983	0.0274	**0.0256**
	Mean	0.0574	0.0574	0.0885	0.0719	0.0403	0.0437	0.0593	0.0337	0.0415	0.1402	0.0244	**0.0221**
RMSE	Soil	0.2688	0.2634	0.2688	0.1833	0.1163	0.0709	0.0707	0.1144	0.1736	0.2318	0.0657	**0.0643**
	Tree	0.2511	0.2534	0.2511	0.1914	0.0750	**0.0392**	0.0454	0.0537	0.2033	0.1066	0.0469	0.0536
	Water	0.4268	0.4262	0.4273	0.1573	0.1189	0.0536	0.0553	0.0772	0.3081	0.0592	0.0555	**0.0186**
	Mean	0.3253	0.3250	0.3255	0.1779	0.1218	0.0561	0.0571	0.0855	0.2355	0.1512	0.0553	**0.0495**
SID		0.0123	0.0123	0.0983	0.0116	0.0191	0.0034	0.0165	0.0028	0.0032	0.4012	0.0025	**0.0018**
AAD		0.6074	0.7075	0.6076	0.2310	0.2148	0.0864	0.0920	0.1178	0.3296	0.2115	0.0736	**0.0645**
AID		5.2738	4.4580	5.2776	1.4970	1.2186	0.2574	0.2883	0.3982	2.3689	1.2936	**0.1274**	0.1823

spectral angle distributions on a unit sphere in Fig. 12, where the proposed framework achieved the lowest SAD score in the most challenging high-noise environment (10 dB). This advantage was sustained at 20, 30, and 40 dB. These results indicate that the proposed architecture mitigates noise interference and preserves intrinsic spectral features. Regarding abundance estimation, WMSM-Net delivered competitive RMSE across noise levels, achieving the lowest RMSE at 20 and 40 dB and remaining within the lowest-RMSE tier at 10 and 30 dB. The abundance maps in Fig. 13 for the 10 dB setting show that, unlike methods that perform well at low noise but degrade at high noise, it delivers the most robust performance. Leading in SAD and maintaining competitive RMSE, WMSM-Net validates its superiority in estimating endmember signatures and abundance maps.

D. Experiments on the Samson Dataset

Table III presents the quantitative unmixing results for the Samson dataset. WMSM-Net achieves the best overall spectral and abundance accuracy. For the additional metrics, WMSM-Net ranks first in SID and AAD and second in AID. This indicates that our model effectively balances spectral accuracy with spatial reconstruction, maintaining high fidelity across all ground materials. Fig. 14 displays the ground truth and the estimated abundance maps. WMSM-Net generates abundance maps with sharper transitions and cleaner details, closely resembling the ground truth. Fig. 15 shows that the endmember signatures estimated by WMSM-Net align almost perfectly with the reference spectra, particularly in the absorption valleys.

E. Experiments on the Jasper Ridge Dataset

The Jasper Ridge dataset presents a more complex scenario with four endmembers. Table IV summarizes the performance of all competing methods. WMSM-Net ranks first in mean SAD and RMSE and also in SID, AAD, and AID. highlighting the effectiveness of the wavelet-enhanced Mamba

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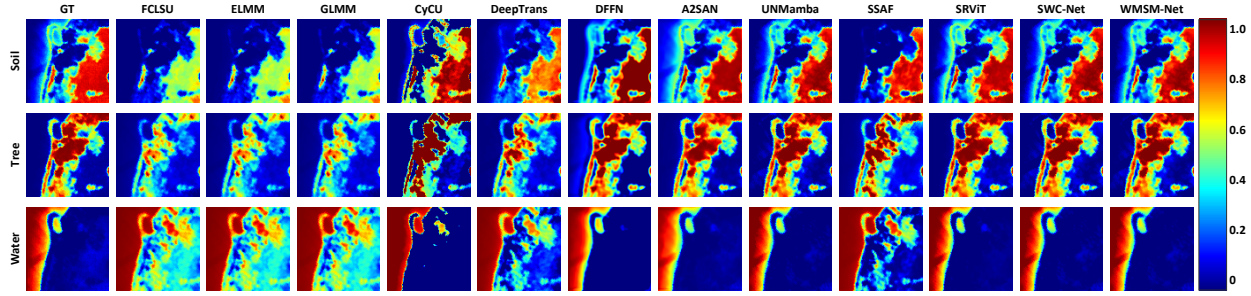


Fig. 14. Visual comparison of the abundance maps estimated by different unmixing methods on the Samson dataset.

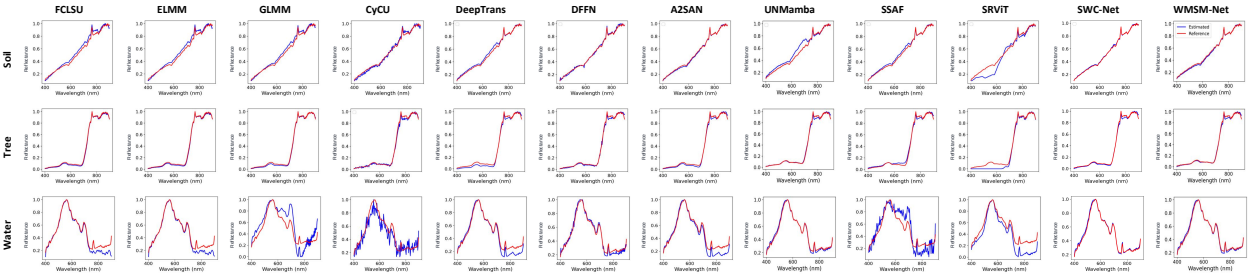


Fig. 15. Visual comparison of the endmembers extracted by different methods on the Samson dataset. Red lines denote the reference spectra, and blue lines denote the estimated spectra. The x-axis represents the spectral wavelength(nm), and the y-axis represents the normalized reflectance.

TABLE IV
QUANTITATIVE PERFORMANCE COMPARISON (SAD, RMSE, SID, AAD, AND AID) ON THE JASPER RIDGE DATASET. THE BEST RESULTS ARE SHOWN IN **BOLD**, AND THE SECOND-BEST RESULTS ARE UNDERLINED.

Methods		FCLSU	ELMM	GLMM	CyCU	DeepTrans	DFFN	A2SAN	UNMamba	SSAF	SRViT	SWC-Net	**WMSM-Net**
SAD	Tree	0.1724	0.1724	0.1687	0.0561	0.0411	0.0764	0.1061	0.0838	**0.0346**	0.0579	0.0487	0.0407
	Water	0.2540	0.2540	0.2578	0.1630	0.0968	0.1129	0.0511	0.0584	0.2216	0.1357	0.0334	**0.0292**
	Dirt	0.1097	0.1097	0.1066	0.0672	0.0515	0.0683	0.0848	0.0659	**0.0257**	0.0507	0.0457	0.0352
	Road	0.0523	0.0523	0.0511	0.1049	0.0436	0.0425	0.1194	0.0585	**0.0253**	0.0275	0.0577	0.0262
	Mean	0.1471	0.1471	0.1460	0.0978	0.0583	0.0750	0.0904	0.0667	0.0768	0.0680	0.0464	**0.0328**
RMSE	Tree	0.1575	0.1576	0.1575	0.0958	0.0484	0.1845	0.0900	0.0620	0.0787	0.0852	0.0971	**0.0303**
	Water	0.1330	0.1332	0.1332	0.1019	0.0599	0.1581	0.1019	**0.0229**	0.1028	0.0704	0.0775	0.0230
	Dirt	0.1122	0.1125	0.1121	0.1403	0.1444	**0.0415**	0.1069	0.0872	0.0799	0.1071	0.1099	0.0666
	Road	0.0776	0.0761	0.0769	0.1188	0.1580	0.1045	0.1195	0.0670	0.0695	0.0937	0.0881	**0.0580**
	Mean	0.1236	0.1235	0.1235	0.1155	0.1137	0.1221	0.1046	0.0642	0.0836	0.0901	0.0932	**0.0481**
SID		0.1203	0.1203	0.1536	0.0238	0.0607	0.0262	1.0802	0.0169	0.0268	0.0198	0.0228	**0.0115**
AAD		0.2122	0.2116	0.2114	0.1726	0.1300	0.1906	1.1483	0.1128	0.1328	0.1555	0.1567	**0.0809**
AID		1.5282	1.5109	1.5240	1.5371	1.9470	0.7481	11.2347	0.5251	0.4927	0.5742	0.6063	**0.3458**

architecture in both endmember extraction and abundance estimation. The abundance maps in Fig. 16 corroborate these findings. The Road endmember is difficult to unmix because of its sparse distribution and spectral similarity to Dirt. Many comparison methods, such as CyCU and DeepTrans, fail to clearly distinguish the road network. However, WMSM-Net reconstructs the Road map with high contrast and continuity. Fig. 17 illustrates the extracted spectral signatures. WMSM-Net remains stable, with the estimated curves tracking the ground truth tightly across the entire wavelength range.

F. Experiments on the Urban Dataset

The Urban dataset is among the most challenging hyperspectral scenarios because of spectrally similar man-made materials and complex spatial distributions. Table V summarizes the performance of all competing methods. WMSM-Net achieves the lowest mean SAD and RMSE and ranks first in SID, AAD, and AID. This indicates that the multi-scale spectral–spatial modeling in WMSM-Net effectively mitigates the spectral confusion common in urban scenes. Qualitatively, Fig. 18 presents the estimated abundance maps. Compared with other methods, WMSM-Net better preserves edges and

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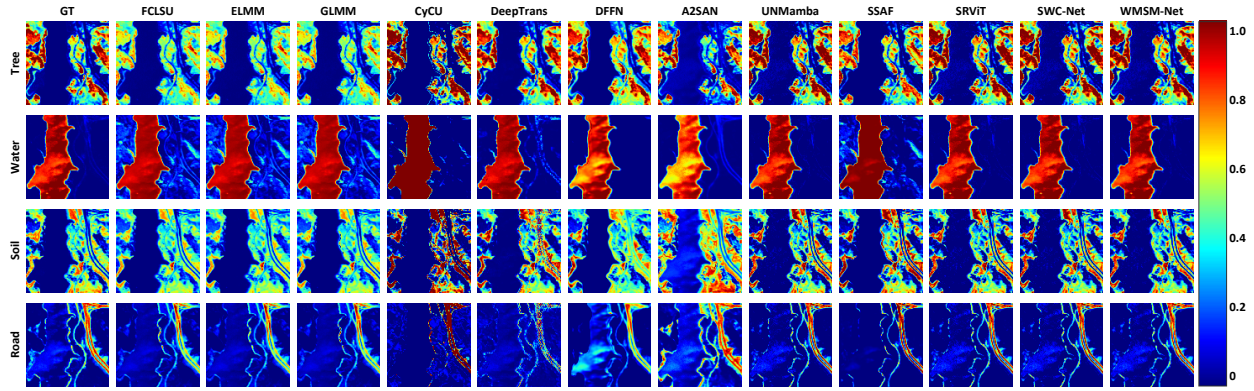


Fig. 16. Visual comparison of the abundance maps estimated by different unmixing methods on the Jasper Ridge dataset.

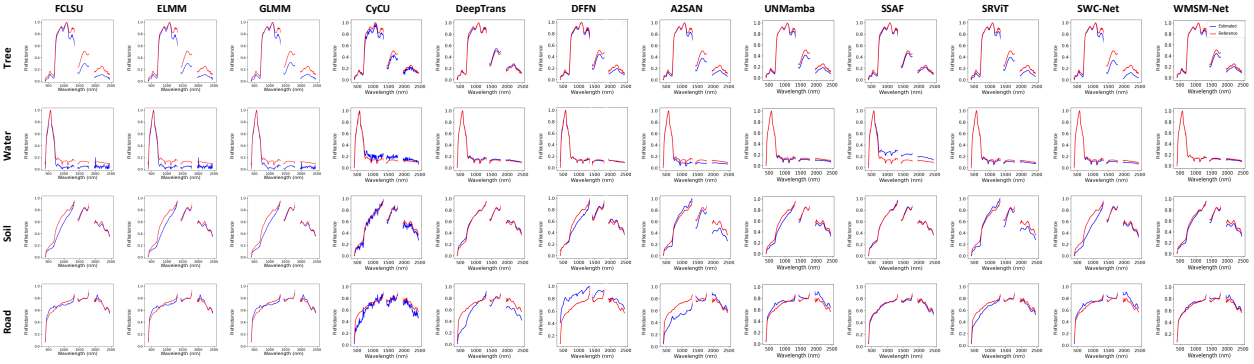


Fig. 17. Visual comparison of the endmembers extracted by different methods on the Jasper Ridge dataset. Red lines denote the reference spectra, and blue lines denote the estimated spectra. The x-axis represents the spectral wavelength(nm), and the y-axis represents the normalized reflectance.

TABLE V
QUANTITATIVE PERFORMANCE COMPARISON (SAD, RMSE, SID, AAD, AND AID) ON THE URBAN DATASET. THE BEST RESULTS ARE SHOWN IN **BOLD**, AND THE SECOND-BEST RESULTS ARE UNDERLINED.

Methods		FCLSU	ELMM	GLMM	CyCU	DeepTrans	DFFN	A2SAN	UNMamba	SSAF	SRViT	SWC-Net	**WMSM-Net**
SAD	Asphalt	0.1316	0.1316	0.1312	0.1130	0.2250	0.1938	0.1013	0.1220	0.1364	0.0990	0.1195	**0.0792**
	Grass	0.5739	0.5739	0.5167	0.1832	0.1259	0.0605	0.0591	0.0560	0.3455	0.1633	**0.0524**	0.0594
	Tree	0.1190	0.1190	0.1179	0.1657	0.1095	0.2593	0.0883	**0.0304**	0.1178	0.0442	0.0525	0.0457
	Roof	0.2174	0.2174	0.2168	0.0999	**0.0359**	0.0678	0.0883	0.0472	0.0954	0.2090	0.1536	0.0431
	Mean	0.2605	0.2605	0.2456	0.1404	0.1241	0.1454	0.0829	0.0638	0.1738	0.1289	0.0945	**0.0568**
RMSE	Asphalt	0.3105	0.3110	0.3105	0.1398	0.1168	0.1102	0.1113	0.0992	0.2938	0.2727	0.2072	**0.0977**
	Grass	0.4836	0.4829	0.4837	0.4008	0.1316	0.1183	**0.0919**	0.1254	0.3644	0.2515	0.3109	0.1095
	Tree	0.2817	0.2817	0.2817	0.2856	0.0892	0.1181	0.0880	**0.0690**	0.2715	0.2729	0.2240	0.0849
	Roof	0.1893	0.1894	0.1893	0.1732	0.1215	0.0838	0.1049	0.0973	0.1476	0.2721	0.1236	**0.0819**
	Mean	0.3337	0.3336	0.3338	0.2701	0.1158	0.1076	0.0990	0.0997	0.2804	0.2675	0.2164	**0.0941**
SID		1.1481	1.1074	0.7161	0.0910	0.1226	0.0475	0.0215	0.0102	0.0521	0.1016	0.0183	**0.0087**
AAD		0.7075	0.7077	0.7076	0.5426	0.2558	0.2363	0.1967	0.2369	0.4934	0.3294	0.4147	**0.1966**
AID		4.4385	4.3954	4.4387	5.7689	1.3808	1.1597	1.0050	0.8603	2.1913	1.9354	1.7333	**0.8198**

internal consistency in its abundance maps. Fig. 19 shows that the Asphalt and Roof spectra contain complex fluctuations that are difficult to reconstruct. Compared with methods exhibiting noticeable jitter and mismatch within certain ranges, WMSM-Net captures these trends accurately. These results confirm that WMSM-Net is robust in handling large-scale, complex urban environments.

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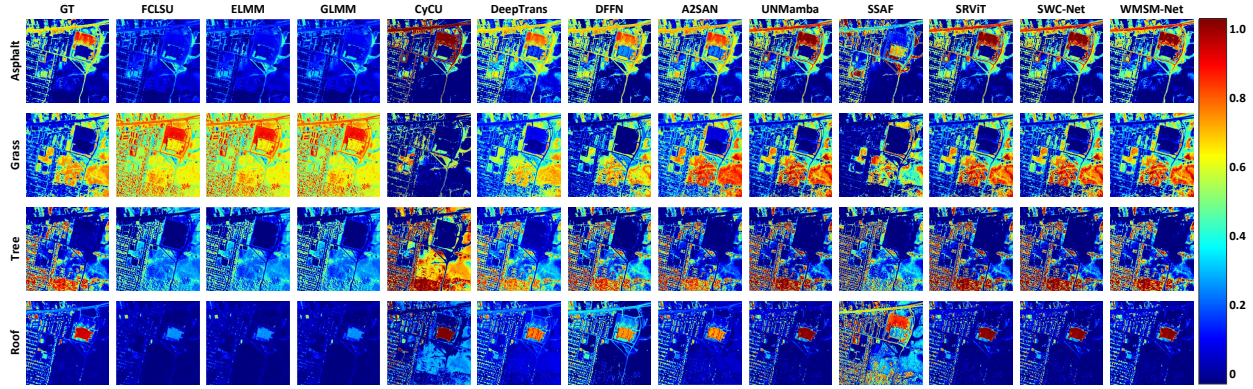


Fig. 18. Visual comparison of the abundance maps estimated by different unmixing methods on the Urban dataset.

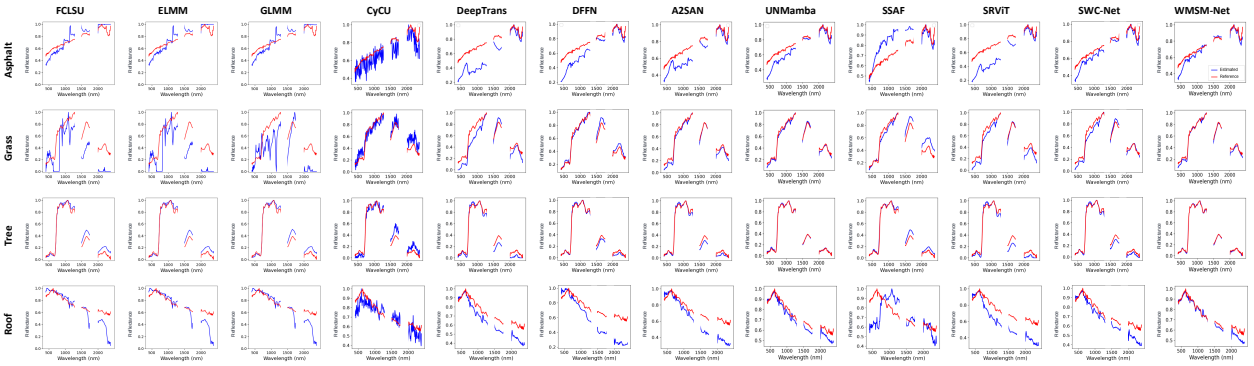


Fig. 19. Visual comparison of the endmembers extracted by different methods on the Urban dataset. Red lines denote the reference spectra, and blue lines denote the estimated spectra. The x-axis represents the spectral wavelength(nm), and the y-axis represents the normalized reflectance.

G. Ablation Experiment

To assess the effectiveness of each component, an ablation study is conducted on the Jasper Ridge dataset, and the results are reported in Table VI. The DWT baseline suffers from a loss of spatial resolution caused by the downsampling operation. Replacing the DWT with the SWT retains the original spatial resolution and leads to a lower RMSE. The FFR rectifies the wavelet sub-bands in the Fourier domain, and the combination of the SWT and the FFR forms the EWT, which yields a structurally consistent representation with suppressed noise for subsequent modeling. The AEM reduces the SAD by modeling the long-range spectral and spatial dependencies, and the AAM fuses the high and low frequency sub-bands through a hierarchical interaction rather than treating them independently. The GCS mechanism refines the estimated endmembers by constraining them to the global spectral distribution of the scene. With the successive incorporation of these components, both SAD and RMSE are progressively reduced, and the full model obtains the lowest values for both metrics, confirming the effectiveness of the proposed design.

H. Computational Cost

Table VII reports the running time, in seconds, of different methods on the Samson, Jasper Ridge, and Urban datasets. The Mamba block of WMSM-Net has linear asymptotic complexity with respect to the sequence length, which provides a scalable mechanism for modeling global dependencies in high-dimensional hyperspectral data. Because the proposed framework includes multi-branch wavelet decomposition, collaborative feature fusion, and physically grounded endmember optimization, which introduce additional computational overhead, the overall processing time of WMSM-Net is not low among the compared methods. These components are designed to enhance frequency-aware feature representation and physically interpretable endmember estimation, thereby improving the accuracy and robustness of hyperspectral unmixing. Therefore, WMSM-Net aims to provide a practical framework for accurate and robust hyperspectral unmixing while maintaining an acceptable computational cost.

V. Conclusion

In this article, we proposed WMSM-Net, a wavelet-driven state-space framework for hyperspectral unmixing. To address the entanglement of spatial textures and sensor noise in the raw spatial-spectral domain, we introduced the EWT, which combines the Stationary Wavelet Transform with the FFR to separate task-relevant spatial textures from high-frequency interference. Building on this frequency-disentangled representation, the CSSMamba encoder exploits the linear complexity of state-space models to efficiently capture long-range spectral-spatial dependencies for accurate abundance estimation. To improve physical fidelity and robustness to spectral variability,

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TABLE VI
ABLATION ANALYSIS OF COMPONENT EFFECTIVENESS ON THE JASPER RIDGE DATASET. THE BEST RESULTS ARE SHOWN IN **BOLD**, AND THE SECOND-BEST RESULTS ARE UNDERLINED.

Wavelet	AEM	AAM	GCS	SAD	RMSE
DWT				0.0520	0.0984
SWT				0.0627	0.0749
EWT				0.0654	0.0665
EWT	✓			0.0422	0.0564
EWT	✓	✓		0.0346	0.0483
DWT	✓	✓	✓	0.0500	0.0756
SWT	✓	✓	✓	0.0354	0.0568
EWT	✓	✓	✓	**0.0328**	**0.0481**

TABLE VII
COMPUTATIONAL EFFICIENCY COMPARISON ON DIFFERENT DATASETS IN TERMS OF SECONDS.

Methods	Samson	Jasper Ridge	Urban
FCLSU	0.1412	0.23	0.46
ELMM	373.18	490.39	490.39
GLMM	237.62	307.68	808.87
CyCU	13.11	2.33	103.04
DeepTrans	20.81	22.82	143.44
DFFN	64.35	44.44	633.08
A2SAN	18.65	13.87	51.28
UNMamba	46.78	22.73	39.77
SSAF	9.47	34.57	36.97
SRViT	139.56	133.78	498.66
SWC-Net	11.46	12.40	128.41
WMSM-Net (Ours)	77.58	60.30	192.52

the Endmember Module employs the Global Correlation Selection mechanism to guide the estimated endmembers toward the global spectral distribution of the scene. Comprehensive experiments on synthetic and real-world datasets demonstrate that WMSM-Net achieves competitive performance compared with existing unmixing methods. Despite these advances, several challenges remain open. Extending the current linear mixing framework to nonlinear scenarios would broaden its applicability to intimate mixtures and multiple-scattering effects, while incorporating spectral variability through augmented or perturbed mixing models [45], [50] may further improve adaptability under complex illumination, atmospheric conditions, and intrinsic material variations. Beyond additive Gaussian noise, enhancing robustness to mixed and structured noise, together with validation on larger-scale and more diverse airborne and satellite benchmarks, are also worth exploring in future work.

APPENDIX

This appendix extends the experimental evaluation to Orchard and Cuprite datasets. Orchard is a physics-based ray-tracing benchmark with reference endmember spectra and fractional-cover maps, enabling quantitative evaluation of both endmembers and abundances. Cuprite is a real-world mineral scene characterized by complex mineral distributions. Together, these datasets provide additional evidence of the proposed method's performance under physically modeled mixing and complex real-world spectral variability.

A. Experiments on the Orchard Dataset

The Orchard dataset represents a citrus orchard generated by physics-based ray tracing. The selected scene contains 20×20 pixels, 216 spectral bands spanning 350–2500 nm, and three endmembers: tree, soil, and weed. Table VIII reports the quantitative comparison. WMSM-Net achieves the lowest MSAD and the lowest MRMSE. As shown in Figs. 20 and 21, the estimated abundance maps retain the main spatial patterns of the reference maps, while the estimated spectra follow the principal trends of the reference signatures. These results demonstrate competitive unmixing performance on a physically generated scene with complex spectral mixing.

B. Experiments on the Cuprite Dataset

The Cuprite dataset represents a challenging hyperspectral unmixing scenario because of its pronounced spectral variability and complex mineral distributions. Captured by the AVIRIS sensor over the Cuprite mining district in Nevada, USA, the selected image has a spatial dimension of 250×190 pixels. After removing noisy and water-vapor absorption bands, 188 valid bands are retained, with spectral wavelengths ranging from approximately 390.09 to 2470.46 nm. The dataset contains 12 reference endmembers: alunite, andradite, buddingtonite, dumortierite, kaolinite 1, kaolinite 2, muscovite, montmorillonite, nontronite, pyrope, sphene, and chalcedony.

Reliable ground-truth abundance maps are unavailable for this scene. Accordingly, we evaluate the results mainly through the estimated endmember spectra and their corresponding abundance maps. As shown in Figs. 23 and 22, WMSM-Net extracts 12 mineral endmembers and produces their spatial abundance distributions. The matched endmembers achieve a mean SAD of 0.0984 rad. The estimated spectra reproduce the overall spectral trends and characteristic absorption features of the reference signatures. Meanwhile, the abundance maps exhibit distinct and spatially coherent mineral distribution patterns. These observations indicate that WMSM-Net can effectively handle spectral variability and complex mineral mixtures in the Cuprite scene.

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TABLE VIII
QUANTITATIVE PERFORMANCE COMPARISON (SAD AND RMSE) ON THE ORCHARD DATASET. THE BEST RESULTS ARE SHOWN IN **BOLD**, AND THE SECOND-BEST RESULTS ARE UNDERLINED.

Methods		FCLSU	ELMM	GLMM	CyCU	DeepTrans	DFFN	A2SAN	UNMamba	SSAF	SRViT	SWC-Net	WMSM-Net
SAD	Tree	**0.2339**	**0.2339**	0.2342	0.2357	0.2401	0.2713	0.2510	0.2608	0.2789	0.2393	0.2503	0.2488
	Soil	0.1568	0.1326	0.1558	0.1464	0.1269	0.1495	**0.1042**	0.1529	0.2274	0.1338	0.1256	0.1283
	Weed	0.1313	0.1148	0.1194	0.1726	0.1400	0.1002	0.1559	0.1701	0.1696	0.1350	0.1807	**0.0969**
	Mean	0.1740	0.1604	0.1698	0.1849	0.1690	0.1736	0.1780	0.1946	0.2253	0.1694	0.1856	**0.1580**
RMSE	Tree	0.4156	0.3867	0.4149	0.6113	0.3915	0.3539	**0.1110**	0.1198	0.1672	0.1560	0.3510	0.1225
	Soil	0.1164	0.1262	0.1160	0.4289	**0.0937**	0.2050	0.1744	0.1682	0.1979	0.1641	0.3656	0.1334
	Weed	0.5119	0.4935	0.5109	0.3940	0.4586	0.4509	0.2292	0.1530	0.1790	0.1698	0.3518	**0.1170**
	Mean	0.3866	0.3692	0.3858	0.4875	0.3523	0.3410	0.1715	0.1470	0.1818	0.1634	0.3565	**0.1244**

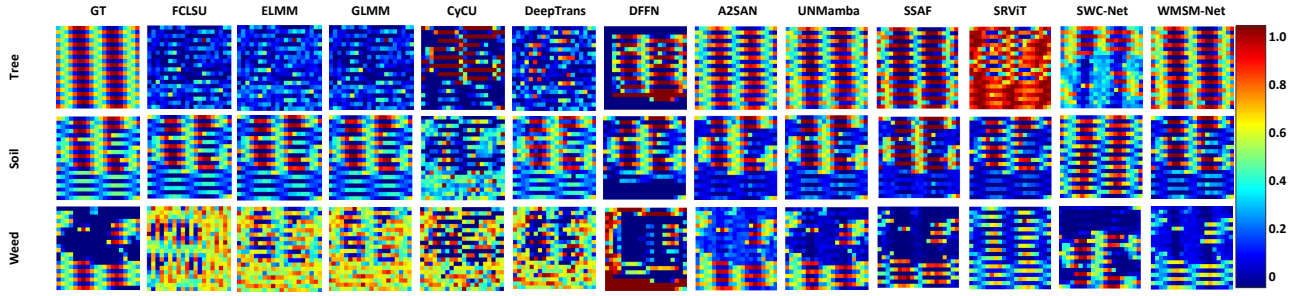


Fig. 20. Visual comparison of the abundance maps estimated by different unmixing methods on the Orchard dataset.

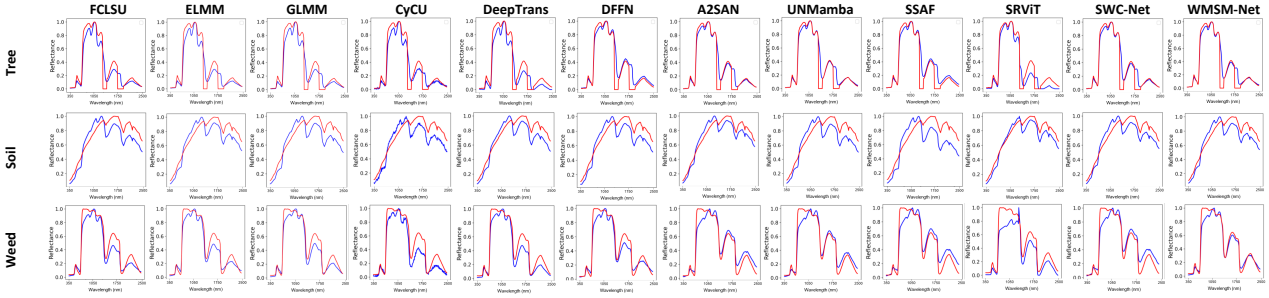


Fig. 21. Visual comparison of the endmembers extracted by different methods on the Orchard dataset. Red lines denote the reference spectra, and blue lines denote the estimated spectra. The x-axis represents the spectral wavelength(nm), and the y-axis represents the normalized reflectance.

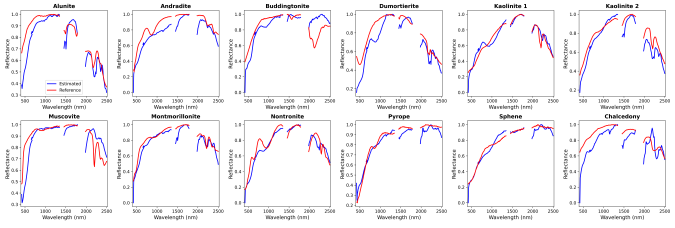


Fig. 22. Visual comparison of the endmembers extracted by WMSM-Net and the corresponding reference library spectra on the Cuprite dataset. Red lines denote the reference spectra, and blue lines denote the estimated spectra. The x-axis represents the spectral wavelength (nm), and the y-axis represents the normalized reflectance.

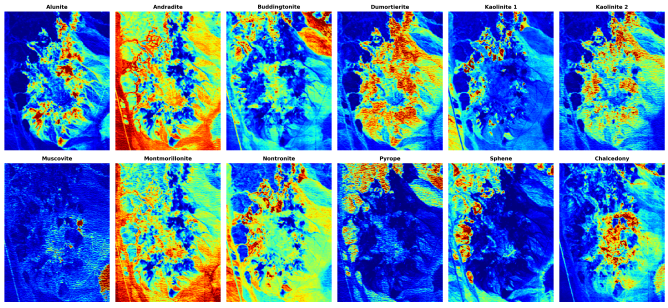


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